SAMPLE

Cosmos Unveiled: Stellar Nucleosynthesis

How Stars Make the Elements

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Introduction

Every atom in your body was assembled in a place with no floor, no walls, and a temperature of twenty million degrees. The calcium in your bones was fused inside a star. The iron in your blood was forged in the last second of a star's life, or in the explosion that ended it. The oxygen you just breathed was manufactured by the nuclear fires of a star that lived and died billions of years ago, before the Sun was born. The gold in a wedding ring may have been born in the collision of two dead city-sized stars, in an event brighter than a billion suns for a few tenths of a second.

The process with the long name, stellar nucleosynthesis, is simply this: the building of chemical elements by nuclear fusion inside stars. Stars are not born with the periodic table ready-made. The universe began with almost nothing but hydrogen and helium, a dash of lithium, and the gravity that would eventually do something with them. Every element beyond those first three, the carbon, the oxygen, the iron, the gold, was created afterward, in the interiors of stars, and was scattered into space when those stars died. Stellar nucleosynthesis is the story of how the raw hydrogen of the early universe became, over billions of years, a universe containing planets, oceans, and people.

The word needs a careful unpacking, because it is a piece of scientific Greek. Nucleosynthesis means the synthesis, the making, of nuclei, the dense cores of atoms. A chemical element is defined by its nucleus: change the nucleus and you change the element. So nucleosynthesis is element-making, and stellar nucleosynthesis is element-making by stars, through nuclear fusion, the welding of light nuclei into heavier ones, with the energy difference paid out as heat and light.

This book tells the story from the ground up. It begins with the abundance puzzle that first made the subject urgent: why the universe's supply of elements is so strangely lopsided, with iron common and gold rare, hydrogen everywhere and beryllium nearly nowhere. It explains how a star works, what allows one nucleus to embrace another when every nucleus in the universe is electrically determined to refuse. It follows a star's life from the hydrogen burning of its youth, through helium, carbon, and the increasingly desperate stages of old age, to the iron core that ends everything, and the explosions and collisions that build the elements beyond.

The people get their chapter here too: the astronomers and physicists who, in a famous series of papers straddling the 1950s, worked out where every element comes from, and settled a question that had stood since the ancients first noticed that the sky and the earth were made of the same things.

And the story is unfinished. The newest elements-heavy events have been watched directly in the last decade, grains older than the Sun have been dug out of meteorites and read like letters from dead stars, and open questions, from missing isotopes to the very first stars, are being worked on right now. Like every book in this series, it deals in real science, told honestly, with the wonder left where it belongs.

A note on scope, so the map is clear from the start. The building of the lightest nuclei, hydrogen, helium, and lithium, belongs to the universe's first minutes and to a companion book in this series; this book owns everything from carbon onward, the work of the stars, and it will touch the early universe only where the two stories share a border. Cosmic rays, the fast particles that slam into the upper atmosphere, also assemble a few light elements in flight, a small side business of the same border region. The heavy elements, and everything they make possible, are the stars' alone, and the chapters ahead walk the whole ladder, from the first helium fusion to the last atom of gold.

Here is how the universe cooked everything you have ever touched.

Chapter 01: The Leftover Universe: What Stars Started With

To understand what stars built, it helps to stand in the emptiest possible place and take inventory. The universe, measured as a whole, is overwhelmingly made of two elements: hydrogen and helium. Roughly nine out of ten atoms are hydrogen, nearly all the rest are helium, and everything else on the periodic table, every other element combined, amounts to a rounding error. The air, the sea, the rocks, the iron core of the Earth, all of that richness is a local specialty, not the cosmic norm.

Astronomers have a bookkeeping term for that richness, and it will recur through this book: metallicity, the fraction of a star's material made of elements heavier than helium. The word is deliberately sloppy, because to an astronomer, carbon, oxygen, and iron are all metals, lumped together for the purpose of measuring how much of a

star's ancestry involved earlier stars. The universe's first gas had essentially zero metallicity; the Sun, born nine billion years later, carries about one and a half percent; and the whole plot of the nucleosynthesis story can be told as the slow, star-by-star rise of that single number. When you read later chapters about astronomers finding the oldest stars, the measurement they are ranking them by is exactly this: the star with the lowest metallicity is the one whose gas has been processed by the fewest earlier furnaces.

That starting point was set in the universe's first minutes. When the cosmos was young, hot, and dense, protons and neutrons welded themselves into the first few nuclei, and the expanding universe cooled before anything heavier than lithium could form in bulk. That early event, described in detail in the companion book of this series on the Big Bang, delivered the raw material: a universe of hydrogen and helium gas, with gravity as the only tool left in the box.

The tool, however, was enough. Gravity is patient and it is relentless. Hydrogen and helium, spread through the cosmos in clouds, began to collect where the density ran slightly higher, and every collection pulled more material toward itself. As the gas fell inward, it heated, and in the centers of the largest collections the temperature crossed the threshold where nuclear fusion ignites. At that moment, the first stars lit up, and the business of building elements began.

It is worth pausing on the strangeness of the raw material. A universe of hydrogen and helium alone would be a universe of gas giants and dark clouds: no rocks, no water, no chemistry beyond a few dull compounds, and certainly no life. Carbon and oxygen, the backbone atoms of biology, simply did not exist. Neither did silicon, which is what planets are mostly made of, nor iron, which is what gives a planet a hot metal heart and a magnetic field. The universe of the first stars was a two-note chord, and everything else, the entire periodic table, would have to be composed, note by note, in the furnaces of stars.

The scale of the transformation is hard to overstate. In the nearly fourteen billion years since, stars have converted about one atom in ten of the primordial hydrogen into helium, and a small fraction of a percent into everything else. The stars are slow, careful cooks: they did not rush the work, and they are nowhere near done.

The scale of the remaining work takes a closing sentence for this chapter. The universe's stars have, in nearly fourteen billion years, converted only a few percent of the primordial hydrogen into anything else, and the great clouds of the cosmos still hold, in their cold hydrogen, the raw material for trillions of years of future stars, if the universe permits them. The periodic table we live among is an early draft. The furnaces that wrote it are still lit, and the book of the elements, begun in the first stars' fires, is very far from its final page. The universe today is still mostly raw material, with the heavy elements concentrated in the inner regions of galaxies, in planets, and in the descendants of the stars that made them.

One number is worth carrying through this whole book, because it explains the shape of everything that follows. Binding energy, the energy holding a nucleus together, peaks at iron. Fusing light nuclei releases energy; fusing nuclei heavier than iron absorbs it. This single fact draws a line through the periodic table and splits the story of nucleosynthesis in two:

Before the theory existed, the puzzle of the abundances was already a science. In the first half of the twentieth century, geochemists measured the elements in Earth's rocks, meteorites, and the Sun's light, and compiled the first rough curve of cosmic abundance, work that crystallized in a famous compilation of the early 1950s that the founders of nucleosynthesis theory carried in their pockets. The curve's features, the hydrogen summit, the iron spike, the sawtooth of even and odd elements, were established facts in search of a machine, and the requirements were strict: whatever built the elements had to reproduce the whole curve, not merely account for a few. The theorists who took up the challenge in the following years were, in the deepest sense, solving the abundance curve, and every chapter of this book is a piece of that solution. everything up to iron can be built by a star during its life, at a profit, and everything beyond iron must be built in dying moments and violent aftermaths, at an energy cost paid by collapse and explosion. Hold that line in mind, and the whole architecture of the subject, the orderly burning stages of a star's life, the iron core that ends it, the explosive and neutron-soaked processes that carry on past iron, falls into place as one connected design.

One piece of vocabulary from the astronomers gets a home early, because it appears in every technical account of this subject and it deliberately scandalizes chemists: the word metal. In stellar astronomy, every element heavier than helium is a metal, all of them together, from carbon to uranium, regardless of chemistry, and a star's proportion of such elements is called its metallicity. The vocabulary is not laziness but compression: for the questions that matter in this book, the sharp chemical distinctions of the laboratory matter far less than the single

act of accounting, how much of a star's mass was assembled in earlier stars and how much is primordial. A metal-poor star is an old star, or at least a star formed from gas that had seen few previous generations, and the metallicity scale that runs across the stellar census is therefore a rough but serviceable chronology of the whole galaxy. The vocabulary also gives this chapter its cleanest sentence: the universe began with a metallicity of essentially zero, and every point of that number since has been earned in the furnaces this book describes.